

Experimental Determination of Moment of Inertia

Alfred Petersson, Jonathan Johansson

Course
Mechanics
Uppsala University

May 11, 2026

Contents

1	Abstract	3
1.1		3
2	Introduction	3
2.1		3
3	Theory/Background	3
3.1		3
4	Method	4
4.1		4
5	Results	5
5.1		5
6	Discussion	5
6.1		5
7	References	5
7.1		5
8	Appendix	5
8.1		5
9	Contributions	5
9.1		5

1 Abstract

1.1

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

2.1

Moment of inertia is a physical property of anything which rotates, characterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate and measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of measuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

3.1

To calculate with rotational movement and cannot idealize the mass to a particle as often done in translational calculations, you have to use expressions giving a link between the mass of an object and how it is distributed I = moment of inertia different geometric forms will have different calculations about its centroid being the center of the distributed mass.

Using this link analogously Newton's second law $\sum F = ma$ can be applied for rotational movement.

$$\sum \tau = I\alpha$$

τ equals torsion and α the angular acceleration

To calculate Torque (τ) it is the distance to where the applied Force perpendicular to the rotational center see figure (fbd)

$$\tau = Tr$$

T is the applied force in Newton and r is the distance to the point of perpendicular applied force

Given an expression for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha$$

In the case that the force is generated with a free falling weight the following substitutions can be made $F = mg = T$ giving the expression

$$mgr = I\alpha$$

torque are given by T times the distance to the perpendicular line the force are applied on

$$Tr = \tau$$

$$I = \frac{mgr}{\alpha} + mr^2$$

4 Method

4.1

$$\sum F = mg - T = ma$$

a = the linear acceleration of the falling mass

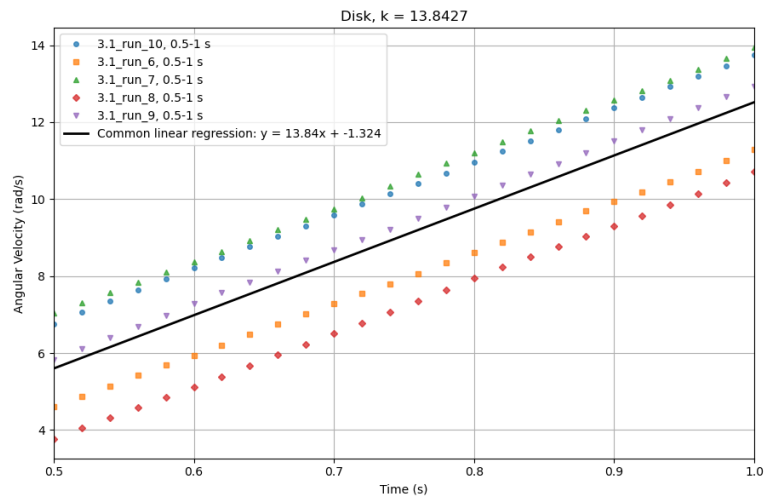


Figure 1: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

5 Results

5.1

6 Discussion

6.1

7 References

7.1

8 Appendix

8.1

9 Contributions

9.1

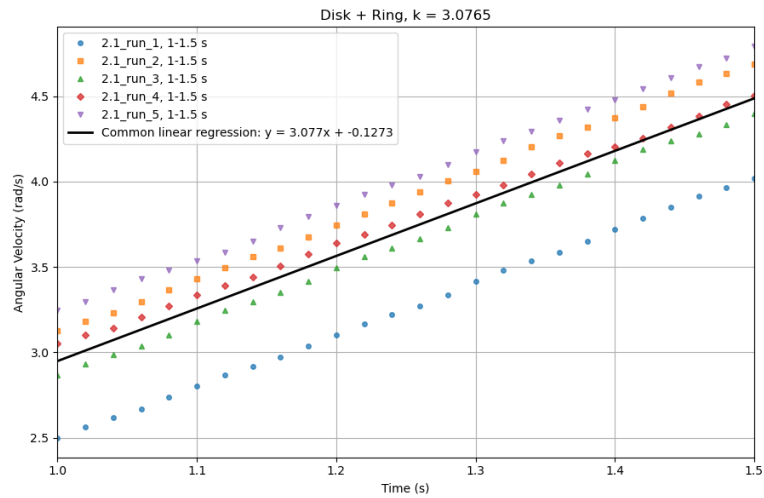


Figure 2: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

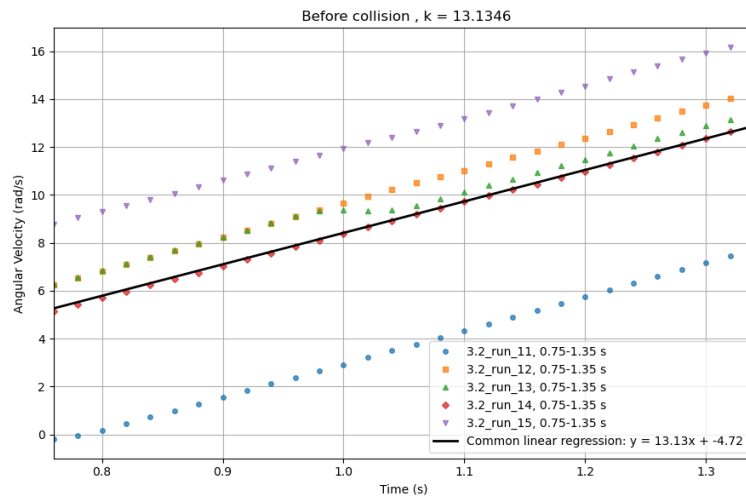


Figure 3: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

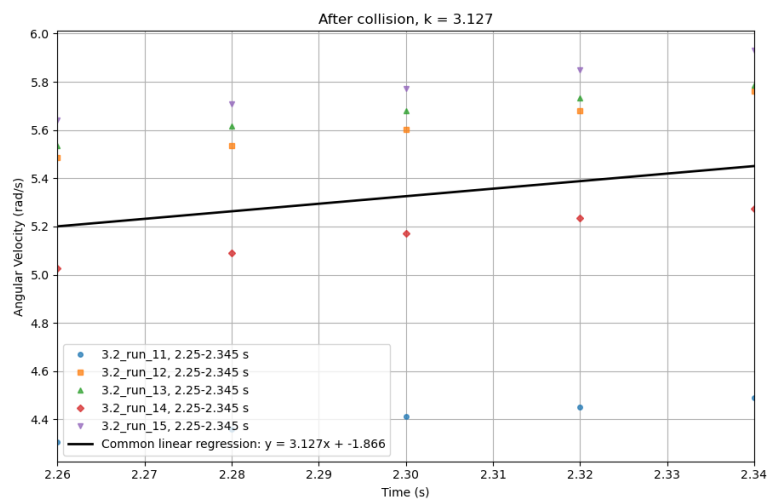


Figure 4: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.